

Niranjan Shrestha

1012-2545 Simcoe St N, Oshawa, ON L1L 0W3, niranjan.stha10@gmail.com, (905) 922-7620

PROFILE SUMMARY

Results-driven and self-motivated Electrical Engineer and Power Electronics Enthusiast with 4+ years of experience in high-frequency/low-frequency power converter design, converter circuit modeling, simulation, control system design, and hardware testing/validation. Technical team lead of the design and end-to-end development of a high-efficiency 7.7 kW wireless charging system for EV with an academic partner. Seeking a suitable role to leverage my knowledge and skills in power electronics design and development projects, circuit modeling and simulation, hardware testing and validation, along with ensuring technical integrity and following proper design procedures and complying with relevant standards.

EDUCATION

Master of Applied Science (Electrical and Computer Engineering) Jan 2022 - Apr 2024
Ontario Tech University • Oshawa, Ontario

- Research area: Wireless charging, power converter and control design

Bachelor's Degree (Electrical Engineering) Nov 2013 - Sep 2017
Tribhuvan University, Pulchowk Campus • Lalitpur, Nepal

- Field of interest: Power electronics, Control system

SKILLS

- Hard skills: MATLAB Simulink/Simscape, M-language, KiCad, LTspice, PLECS, TI microcontroller, STM32, Proteus Design Suite, Python, C/C++ programming, AutoCAD, COMPASS, MS Office, Technical writing, WPT standards (SAE, CSA, UL), single-line diagrams, RT-LAB, OPAL-RT, SPI, UART communication
- Soft skills: Quick learning, analytical problem-solving, critical thinking, team player, communication, initiative and self-motivation, leadership, and adaptability

WORK EXPERIENCE

Electrical Engineer May 2024 - Present
Wireless PnC Inc. • Victoria, British Columbia

- Design, modeling, simulation (circuit level and system level), control, and development of power converters for wireless power transfer systems for E-scooters, E-bikes, and Evs (up to 7.7 kW).
- Conducting research on novel converters and controls for wireless charging systems.
- Generating budgets and time estimates for the development of new prototypes.
- Mentoring and supervising electrical interns for simulation, hardware design, end-to-end development, and test validation for a 7.7 kW wireless charging system.
- PCB design and fabrication of different converters (Boost PFC rectifier, Inverter, etc).
- Collecting information on operational performance, performing root-cause analysis of identified faults, and fixing them.
- Maintaining appropriate design documentation and using company-designated design guidelines, and keeping up to date with new standards in the automotive industry
- Writing validation test plans for all phases of the projects and supporting vehicle-level bench testing activities and reporting findings.

Graduate Research Assistant

Jan 2022 – Apr 2024

Ontario Tech University • Oshawa, Ontario

- Completion of end-to-end R&D of a high-efficiency resonant WPT system (incorporating novel compensation network and phase shift control for CC/CV charging) funded by NSERC and MITACS in collaboration with an industry partner
- Design, modeling, simulation, control, and development of power converters, including DC-DC converters, inverters, rectifiers, and resonant converters, particularly applicable to wireless power transfer systems for EVs.
- Experimental testing and validation of WPT systems with various coil configurations and compensation circuits.

Electrical Engineering Intern

Jun 2022 - Apr 2023

Oshawa Power • Oshawa

- Creating regression models to forecast energy production and revenue generation from several solar PV generation systems and wind turbine generation systems, along with formulating its detailed analysis report.
- Assist in preparing micro-embedded generation connection applications for the net metering system.
- Reviewing and approving major and non-major equipment based on the compliance of their test reports according to the corresponding industry standards.

Project Engineer

Nov 2019 - Dec 2021

Sunshine Energy Pvt. Ltd. • Kathmandu, Nepal

- Site survey along with overall system design, cost estimation, and tender bid document preparation of off-grid and on-grid solar PV system, solar water pumping/irrigation system, solar home system, and solar street lighting system projects.

Electronics Technician

Feb 2018 - Aug 2019

Defense Infrastructure Organization - British Gurkhas Nepal • Lalitpur, Nepal

- Carry out the quality diagnosis and fault-finding on electrical equipment like solar water heaters, generators, motors, and boilers, along with their necessary maintenance and servicing.
- Execute all the necessary testing & inspection of the electrical installation as per the British Standard (BS 7671:2008), including correct documentation and report submission.

VOLUNTEER WORK**Active volunteer**

May 2022 - Present

IEEE Toronto - Durham Region Sub-section • Oshawa, Ontario

Volunteer

Mar 2022 - Mar 2023

Spring open house - UOIT • Oshawa

Event Organizer/Volunteer

Jan 2017 - Feb 2017

Energy Hackathon • Lalitpur, Nepal

AWARDS AND HONOURS

IEEE ITEC Student Travel Award Recipient

2023

ITEC Coil Winding Competition Winner (1st Runner-up)

2023

IEEE SPEC Scholarship Award

2022

President's Appreciation Award (Kathmandu Valley Leo Club)

2016

REFERENCE:

Available upon request